Health and disease

The human body looks after itself without much thought, or, to quote from one syllabus, *'without any conscious intervention'*. There are many things that the human body does without having to think about it – the autonomic system controls many of the involuntary processes that keep us alive.

A dictionary definition of health is *'the condition of being sound in body, mind, or spirit; especially freedom from physical disease or pain'*. A disease is anything that impairs (makes worse) how an organism functions. Most diseases are recognisable by what they do to an organism and the symptoms they cause. Doctors often diagnose an illness using the symptoms it causes.

1 What is meant by *intervention*?

2 Name five things that the body does without thinking to ensure survival.

3 Give another definition of *health*.

4 Give another definition of *disease*.

5 What is meant by *symptom*?

6 Define *diagnose*.

Health is maintained in the human body by a range of systems including:

- genes – the units of inheritance. They carry on specific characteristics from generation to generation and can pass on traits (qualities) that help an organism fight off infection.
- mitosis – a form of cell division that results in genetically identical copies of somatic (non sex) cells. This process replaces dead and injured cells. White blood cells (leukocytes) are cells that resist invasion by foreign organisms and are produced by mitosis in large numbers when an infection occurs.
- cell differentiation – when an immature cell (resembling many other cells) changes to a specific mature cell.
- cell specialisation – means that individual cells carry out specific roles; some of these roles involve the maintenance of health.

(Note: Remember the order of increasing complexity: cells / tissues / organs / organ system / body.)

7 Name two other characteristics that genes could carry.

8 Meiosis is another type of cell replication. Outline how this differs from mitosis.

9 Name another type of blood cell apart from white blood cells.

10 What is meant by an 'immature' cell?

Diseases

A disease that can be caught from another living creature is described as *infectious*. If it can't be transmitted it is called *non-infectious*. An infectious disease can be caused by:

- prion (**p**rotinaceous + **i**nfectious + **on**)
- bacteria (singular *bacterium*)
- viruses
- fungi (singular *fungus*)
- protozoa
- macro parasites

11 Name ten different diseases that affect humans.

12 Give a symptom of each disease.

13 Divide your list of diseases from question 11 into infectious and non-infectious.

14 Give a definition for *microbe*.

Pathogens

A pathogen is an organism that causes a disease when it lives in or on another organism. Pathogens can cause the following effects when they infect an organism:

- destroy cells
- produce toxins (poisons)
- interfere with the normal functioning of the organism
- cause an immune response (producing antibodies and lymphocytes) that, if excessive, harm the organism.

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Causes	Description	Diseases
Prions	Tiny protein particles without nucleic acid, thought to cause various infectious diseases of the nervous system.	scrapie, Creutzfeldt-Jakob disease
Viruses	Sub-microscopic, but complex molecules, that typically contain a protein coat surrounding an RNA or DNA core of genetic material but no semipermeable membrane. Capable of growth and multiplication only in living cells.	chicken pox, hepatitis, influenza, measles, polio, rabies
Bacteria	Microscopic, unicellular organisms.	cholera, pneumonia, tuberculosis
Protozoa	Single-celled parasites, which resemble animals and may form large colonies.	malaria, yellow fever
Fungi	Single or multicelled organisms that produce spores and obtain food directly by absorption. Sometimes classified as plants without chlorophyll.	ringworm, athlete's foot
Macro parasites	Small organisms that live on other organisms and cause them harm, eg ticks, fleas, flukes (a type of flatworm).	schistosomiasis or bilharziasis (a widespread disease caused by the infestation of the human body by flukes)

For a pathogen to infect an organism it must be:

- able to survive the transmission from one host to another.
- able to enter the host.
- virulent – able to overcome body defences and only need to be initially present in small numbers.

One defence barrier to prevent entry of pathogens in humans is skin.

15 Identify three effects a pathogen can cause when it infects an organism.

16 Identify two other external defence barriers against pathogens present in humans.

Antigens

In addition to external barriers such as skin, mucous membranes and cilia, the human body has internal defences against pathogens. These include antigens, molecules that trigger the immune response. Antigens are molecules that invade the body. The immune system recognises them as foreign and produces antibodies to bind to them, thus marking them for destruction. After antibodies have been produced, people are often described as having built up an immunity to a particular illness.

Inflammation is a complex defence mechanism, where the body sends greater amounts of blood and lymph to the affected area. The symptoms of inflammation include swelling, tenderness, increased temperature and redness. Inflammation is a reaction by a part of the body to injury, infection, trauma or environmental conditions.

An injury resulting in tissue damage may be caused by a cut, knock, sprain or repetitive strain injury. Infections are caused by pathogens such as bacteria, fungi and viruses or by the toxins they produce. Trauma can be described as either physical injury or nervous shock experienced by a person. Environmental conditions that cause inflammation may contain allergens (substances that cause allergies) such as animal fur, dust, mites, moulds, pollen, smoke or harsh chemicals.

17 Describe what is meant by *immunity*.

18 Explain what is meant by a *pathogen*.

19 Explain what is meant by a *toxin*.

20 What is *lymph*?

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